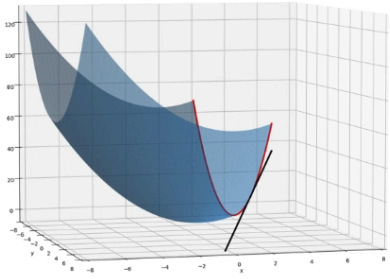
Gradient

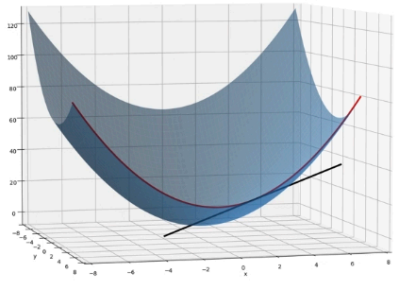
$$f(x, y) = x^2 + y^2$$

Treat y as a constant



$$\frac{\partial f}{\partial x} = 2x$$

Treat x as a constant

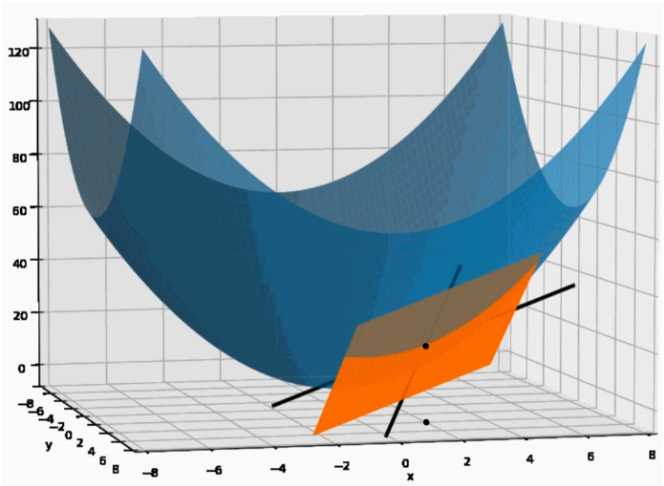


$$\frac{\partial f}{\partial y} = 2y$$

Gradient

$$\begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$



$$f(x, y) = x^2 + y^2$$

The gradient of $f(x, y)$ is: $\nabla f = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$

TASK

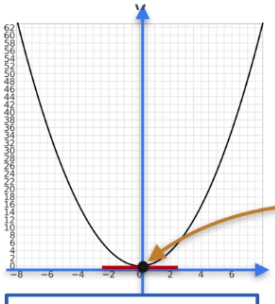
Find the gradient of $f(x, y)$ at (2,3)

The gradient of $f(x, y)$ is given as:

$$\nabla f = \begin{bmatrix} 2 \cdot 2 \\ 2 \cdot 3 \end{bmatrix}$$

Functions of two variables

$f(x) = x^2$



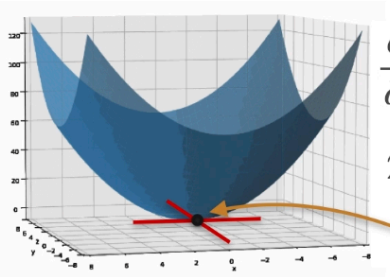
$$f'(x) = 0$$

$$2x = 0$$

$$x = 0$$

Minimum is when slope = 0

$f(x, y) = x^2 + y^2$



$$\frac{\partial f}{\partial x} = 0 \text{ and } \frac{\partial f}{\partial y} = 0$$

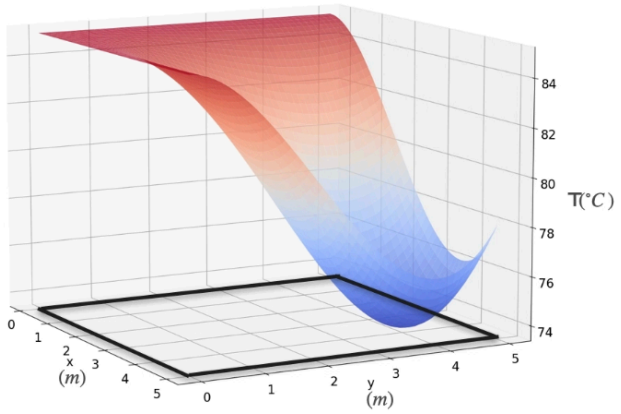
$$2x = 0 \text{ and } 2y = 0$$

$$(x, y) = (0,0)$$

Minimum is when both slopes = 0

Example: Optimization using the gradient

$$T = f(x,y) = 85 - \frac{1}{90}x^2(x-6)y^2(y-6)$$



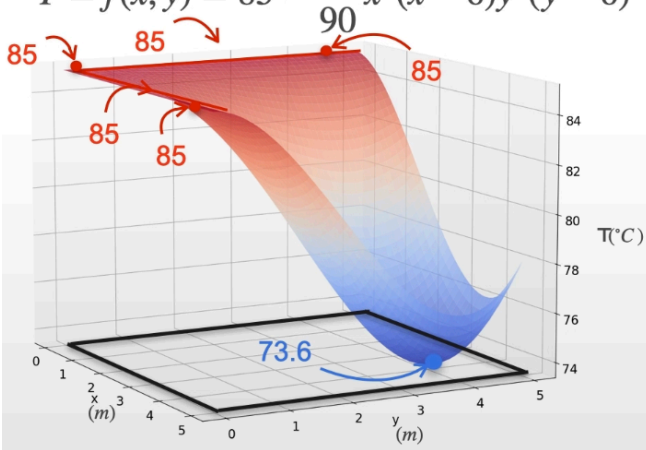
$$\frac{\partial f}{\partial x} = -\frac{1}{90}\boxed{x}\boxed{(3x-12)}\boxed{y^2}\boxed{(y-6)} = 0$$

$x = 0$ $\quad x = 4$ $\quad y = 0$ $\quad y = 6$

$$\frac{\partial f}{\partial y} = -\frac{1}{90}\boxed{x^2}\boxed{(x-6)}\boxed{y}\boxed{(3y-12)} = 0$$

$x = 0$ $\quad x = 6$ $\quad y = 0$ $\quad y = 4$

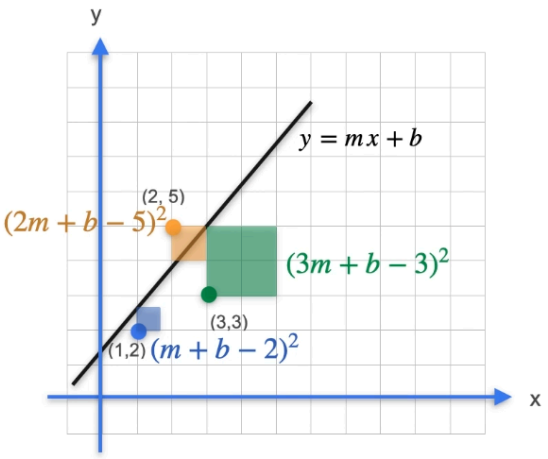
$$T = f(x,y) = 85 - \frac{1}{90}x^2(x-6)y^2(y-6)$$



Candidate points for the minima

$x = 0$, $y = 0$	Maxima
$x = 0,\ y = 0$; $x = 0,\ y = 4$	Maxima
$x = 0,\ y = 6$	Outside
$x = 4,\ y = 0$	Maxima
$x = 4,\ y = 4$	Minimum
$x = 6,\ y = 0$; $x = 6,\ y = 6$	Outside

Analytical Approach

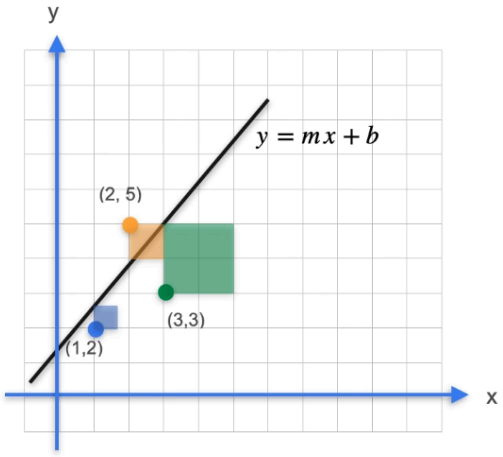


Goal: Minimize sum of squares cost

$$(m + b - 2)^2 \ + \ (2m + b - 5)^2 \ + \ (3m + b - 3)^2$$

$$m^2 + b^2 + 4 + 2mb - 4m - 4b$$
$$+4m^2 + b^2 + 25 + 4mb - 20m - 10b$$
$$+9m^2 + b^2 + 9 + 6mb - 18m - 6b$$

$$E(m,b) = 14m^2 + 3b^2 + 38 + 12mb - 42m - 20b$$



Goal: Minimize sum of squares cost

$$E(m,b) = 14m^2 + 3b^2 + 38 + 12mb - 42m - 20b$$

$$\frac{\partial E}{\partial m} = 28m + 12b - 42$$

$$\frac{\partial E}{\partial b} = 6b + 12m - 20$$

Goal: Minimize sum of squares cost

$$E(m,b) = 14m^2 + 3b^2 + 38 + 12mb - 42m - 20b$$

$$\frac{\partial E}{\partial m} = \boxed{28m + 12b - 42} = 0$$

$$\frac{\partial E}{\partial b} = \boxed{6b + 12m - 20} = 0$$

$m =$

$b =$?

$$12b + 24m - 40 = 0$$

$$4m - 2 = 0$$

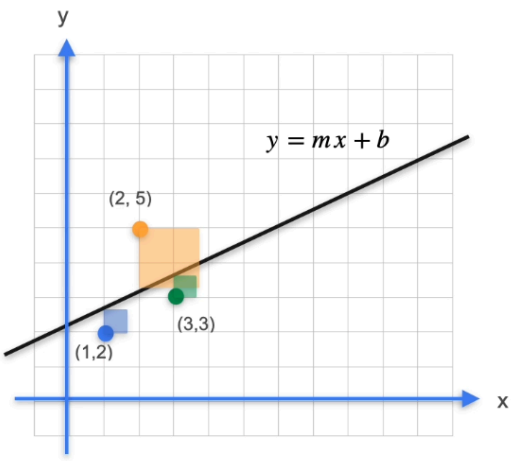
$$m = \frac{2}{4} = 0.5$$

$$6b + 12(0.5) - 20 = 0$$

$$6b + 6 - 20 = 0$$

$$6b - 14 = 0$$

$$b = \frac{14}{6} = \frac{7}{3}$$



$$m = \frac{1}{2}$$

$$b = \frac{7}{3}$$

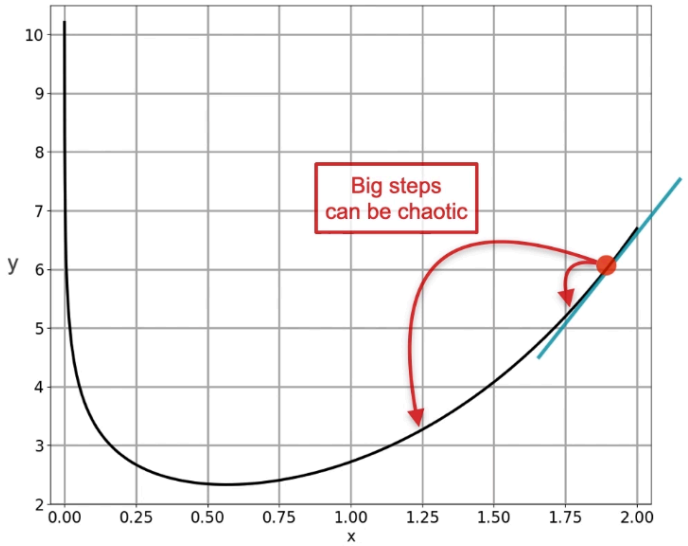
$$E(m = \frac{1}{2}, b = \frac{7}{3}) \approx 4.167$$

Method 2: Be Clever

Try something smarter…

new point = old point - slope

$$x_1 = x_0 - f'(x_0)$$

$$x_1 = x_0 - \alpha \, f'(x_0)$$

Learning rate